TEST YOUR UNDERSTANDING

For more multiple-choice questions, go to the **Practice Test** in the **Mastering Biology** eText or Study Area, or go to goo.gl/GruWRg.

Levels 1-2: Remembering/Understanding

1. Choose the pair of terms that correctly completes this sentence: Catabolism is to anabolism as ___________ is to ___________.
(A) exergonic; spontaneous
(B) exergonic; endergonic
(C) free energy; entropy
(D) work; energy

2. Most cells cannot harness heat to perform work because
(A) heat does not involve a transfer of energy.
(B) cells do not have much thermal energy; they are relatively cool.
(C) temperature is usually uniform throughout a cell.
(D) heat can never be used to do work.

3. Which of the following metabolic processes can occur without a net influx of energy from some other process?
(A) ADP + Ⓟ$_i$ + → ATP + H_2O
(B) $C_6H_{12}O_6$ + 6 O_2 → 6 CO_2 + 6 H_2O
(C) 6 CO_2 + 6 H_2O → $C_6H_{12}O_6$ + 6 O_2
(D) Amino acids → Protein

4. If an enzyme in solution is saturated with substrate, the most effective way to obtain a faster yield of products is to
(A) add more of the enzyme.
(B) heat the solution to 90°C.
(C) add more substrate.
(D) add a noncompetitive inhibitor.

5. Some bacteria are metabolically active in hot springs because
(A) they are able to maintain a lower internal temperature.
(B) high temperatures make catalysis unnecessary.
(C) their enzymes have high optimal temperatures.
(D) their enzymes are completely insensitive to temperature.

Levels 3-4: Applying/Analyzing

6. If an enzyme is added to a solution where its substrate and product are in equilibrium, what will occur?
(A) Additional substrate will be formed.
(B) The reaction will change from endergonic to exergonic.
(C) The free energy of the system will change.
(D) Nothing; the reaction will stay at equilibrium.

Levels 5-6: Evaluating/Creating

7. **DRAW IT** Using a series of arrows, draw the branched metabolic reaction pathway described by the following statements, and then answer the question at the end. Use red arrows and minus signs to indicate inhibition.

L can form either M or N.

M can form O.

O can form either P or R.

P can form Q.

R can form S.

O inhibits the reaction of L to form M.

Q inhibits the reaction of O to form P.

S inhibits the reaction of O to form R.

Which reaction would prevail if both Q and S were present in the cell in high concentrations?
(A) L → M
(B) M → O
(C) L → N
(D) O → P

8. **EVOLUTION CONNECTION** Some people argue that biochemical pathways are too complex to have evolved because all intermediate steps in a given pathway must be present to produce the final product. Critique this argument. How could you use the diversity of metabolic pathways that produce the same or similar products to support your case?

9. **SCIENTIFIC INQUIRY • DRAW IT** A researcher has developed an assay to measure the activity of an important enzyme present in pancreatic cells growing in culture. She adds the enzyme's substrate to a dish of cells and then measures the appearance of reaction products. The results are graphed as the amount of product on the *y*-axis versus time on the *x*-axis. The researcher notes four sections of the graph. For a short period of time, no products appear (section A). Then (section B) the reaction rate is quite high (the slope of the line is steep). Next, the reaction gradually slows down (section C). Finally, the graph line becomes flat (section D). Draw and label the graph, and propose a model to explain the molecular events occurring at each stage of this reaction profile.

10. **WRITE ABOUT A THEME: ENERGY AND MATTER** Life requires energy. In a short essay (100–150 words), describe the basic principles of bioenergetics in an animal cell. How are the flow and transformation of energy different in a photosynthesizing cell? Include the role of ATP and enzymes in your discussion.

11. **SYNTHESIZE YOUR KNOWLEDGE**

Explain what is happening in this photo in terms of kinetic energy and potential energy. Include the energy conversions that occur when the penguins eat fish and climb back up on the glacier. Describe the role of ATP and enzymes in the underlying molecular processes, including what happens to the free energy of some of the molecules involved.

For selected answers, see Appendix A.